

Group Project: Foundations of AI for Business (FAB): Building a Working AI System from First Principles

Welcome to the FAB Group Project. The project goal is to prototype a working AI system from first principles (foundational AI ideas) that solves a problem with potential business value.

- AI “system” does *not* mean just an AI app or tool. It should be a *carefully-architected set of AI ideas and technologies* that *coherently* addresses the fundamental credit assignment challenges specific to the problem. Your group will build this system week-by-week, as we go through the foundational AI ideas of the course.
- Though the project will call on your group to use LLMs throughout the weekly modules, the ultimate AI system that you build may or may not rely on LLMs. Indeed, your success in this project depends on your ability to think critically about how any diverse AI ideas relate to building your AI system coherently out of many parts.

Groups | 3-4 students (preferred). To encourage learning together as a class, all groups will be asked to build their AI system to solve the same problem.

Problem | Designing an AI System for Studying AI Ideas:

You are a group of students with a common problem: employers now expect you to have a deep understanding of AI, yet you have limited processing capacity to read all the information available about AI.

How can you design an “AI system for studying AI ideas” that meets the following criteria: (1) helps you economize on information processing needed to understand AI ideas; yet (2) still enables you to understand these ideas deeply?

Expected Output | A prototype (a piece of software) + a short (2 minute) video or in person presentation of a working system for addressing the above criteria. A prototype is *not* a complete product, but it is also *not* just a concept or a wireframe. For this course, a prototype simply means that we (your professor and students) can easily witness and understand how your system actually functions. You may use any software tools that allow you to build your prototype. No need to use paid tools. No requirement that group members be expert coders.

Learning Objectives | The Project has two goals for your own personal development:

- (1) Long-run: Hands-on experience of the AI ideas from class, enabling a robust understanding of AI foundations that remain stable even as AI technologies change during your career;
- (2) Short-run: Acquiring a currently hot skill that employers want from — building AI systems based on deeply understanding and carefully defining the underlying problem, and not just by adding features to currently-popular tools.

Structure | Our strategy will be to decompose the problem of building your AI system into weekly project modules, each of which will relate to that week’s foundational AI idea.

- **Search**: Defining an initial state, goal state, operators, heuristics.

- **Evolution**: Defining a contribution function and landscape for your goal state.
- **Representation**: Defining various constraints on your data.
- **Knowledge representation and reasoning**: Developing ways of analyzing your data.
- **Critical technical practice**: Defining the language you use to describe your system.
- **Interaction**: Designing an architecture for interacting with your system.
- **Constraint satisfaction**: Defining a constraint language / decision-making architecture.
- **Machine learning**: Optimizing your AI system based on defining KPIs/OKRs.
- **Distributed problem-solving**: Designing a communication and coordination system.
- **Cellular automata**: Defining a scalable replication strategy and data architecture.
- **Business model**: Defining a scalable business model.

Grading

To incentivize all of us to work together as a class — with groups freely sharing their ideas each week — the baseline evaluation for the Group Project will be **pass/fail**. The pass/fail criteria will be that your group's prototype basically reflects the spirit of the AI ideas that we cover in class.

However, each group member will be asked to fill out a detailed **peer evaluation**, including evaluating whether you have shown up on time for and contributed to weekly project modules. Hence, punctual attendance and active participation are essential to your individual grade for the project.

As further incentive, the **top three projects** from the class will be selected for **Demo Day** during the final week of class, with Group members receiving two extra credit points towards their overall course grade (contingent on the peer evaluation). On Demo Day, each of the selected groups will present their prototype, and a **Demo Day Winner** will be selected (no additional extra credit).